CSE 270 Sports Analytics

Predicting with models

Habet Madoyan

Poisson regression

Predicting game outcome with Poisson regression

Poisson regression

Libraries used

```
library(dplyr)
library(VGAM)
library(ggplot2)
library(SportsAnalytics270)
library(BradleyTerry2)
```

Poisson regression

First we are going to test the hypothesis that Home goals and Away goals are independent.

We will use a χ^2 test of independence of two categorical variables

$$H_0 : \text{Two variables are independent}$$

$$H_1 : \text{Two variables are NOT independent}$$

Calculate p-value and compare with significance level of α, if p-value $< \alpha$ reject the NULL hypothesis

Poisson regression

First we are going to create categoric variables for Home and Away team goals in a way that more than 3 goals will be combined together

```
data("f_data_sm")
df = f_data_sm %>% mutate(FTAG = ifelse(FTAG>=4, "4+", FTAG),
                          FTHG = ifelse(FTHG >=4, "4+", FTHG))
```

Poisson regression

Recall that χ^2 test is using Expected and Observed frequencies as an input. Thus we will need to provide a table with joint frequencies of Home and away team goals

```
table(Away=df$FTAG, Home = df$FTHG)
```

```
##      Home
## Away     0     1     2     3    4+
##    0  7794 10250  8035  4395  3101
##    1  6878 11512  8530  4440  2888
##    2  4367  6029  4816  2332  1417
##    3  1925  2646  1693   916   511
##    4+ 1161  1288   828   342   164
```

Poisson regression

We do the test for the last season

- We use function chisq.test
- We specify table with Observed Frequencies
- And specify simulate.p.values = TRUE: This will add robustness to our test
- We do the test for each country and extract p-values of the test

Poisson regression

```
df %>% filter(SEASON==2022) %>% group_by(COUNTRY) %>%
  summarise(pvalue=chisq.test(table(FTAG, FTHG),
                              simulate.p.value=T)$p.value)
```

```
## # A tibble: 11 x 2
##    COUNTRY     pvalue
##    <chr>        <dbl>
##  1 Belgium     0.118
##  2 England     0.0535
##  3 France      0.351
##  4 Germany     0.0355
##  5 Greece      0.222
##  6 Italy       0.681
##  7 Netherlands 0.116
##  8 Portugal    0.386
##  9 Scotland    0.757
## 10 Spain       0.567
## 11 Turkey      0.688
```

The hypothesis of independence is not rejected, at $\alpha = 0.05$, Thus they are independent

Poisson regression

- Poisson regression is used when the dependent variable is a count data, i.e. y=0,1,2,3. . . , thus we assume that the dependent variable has Poisson distribution
- Independent variables can be both numeric and categorical
- Poisson regression is a part of Generalized Linear Regression (GLM) models family

Poisson regression: the general definition

In this chapter we are fitting the model of the following form

$$log(\lambda) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + ...$$

Where λ is the mean of Poisson distribution and $X_1, ..X_n$ are the independent variables

With the estimate of the λ we can generate Poisson distribution and calculate the probabilities for the number of the goals

Preparing data for the Poisson regression model

The initial data for Portugal, season 2021-2022

Making Predictions for Portugal Liga 1

```
data(f_data_sm)
portugal = f_data_sm %>%
            filter(SEASON==2022, COUNTRY=="Portugal") %>%
            select(HOMETEAM, AWAYTEAM, FTHG, FTAG)

head(portugal, n=5)
```

```
##           HOMETEAM     AWAYTEAM FTHG FTAG
## 689712  Sp Lisbon       Vizela    3    0
## 689812     Arouca      Estoril    0    2
## 689912 Moreirense      Benfica    1    2
## 690012   Maritimo     Sp Braga    0    2
## 690112  Guimaraes Portimonense    0    1
```

```
dim(portugal)
```

```
## [1] 306   4
```

Preparing data for the Poisson regression model

The data need to be in the following format: 2 rows for each game and the indicator showing if the team is a home team

```
port1=data.frame(portugal[,c("HOMETEAM", "AWAYTEAM", "FTHG")], Home=1)
port2=data.frame(portugal[,c("AWAYTEAM", "HOMETEAM", "FTAG")], Home=0)
colnames(port1)=c("Team", "Opponent", "Goal", "Home")
colnames(port2)=c("Team", "Opponent","Goal", "Home")
portugal2=rbind(port1, port2)
head(portugal2)
```

```
##              Team     Opponent Goal Home
## 689712  Sp Lisbon       Vizela    3    1
## 689812     Arouca      Estoril    0    1
## 689912 Moreirense      Benfica    1    1
## 690012   Maritimo     Sp Braga    0    1
## 690112  Guimaraes Portimonense    0    1
## 690212    Tondela  Santa Clara    3    1
```

Preparing data for the Poisson regression model

Now let's have a look at the first game of the season: Sporting Lisbon vs Vizela

- Sporting Lisbon was the home team, and the game ended with 3-0

```
portugal[1,]
```

```
##         HOMETEAM AWAYTEAM FTHG FTAG
## 689712 Sp Lisbon   Vizela    3    0
```

- Let's look at our new dataset

```
portugal2[1,]
```

```
##             Team Opponent Goal Home
## 689712 Sp Lisbon   Vizela    3    1
```

```
portugal2[307,]
```

```
##           Team  Opponent Goal Home
## 6897121 Vizela Sp Lisbon    0    0
```

The number of the rows in the portugal2 dataset is twice of the original one

```
dim(portugal)
```

The model

```
model=glm(Goal~Team+Opponent+Home, data=portugal2, family=poisson(link=log))
coefficients(model)
```

```
##           (Intercept)        TeamBelenenses           TeamBenfica
##          -0.032098899          -0.265046266           0.927042742
##           TeamBoavista           TeamEstoril         TeamFamalicao
##           0.260554198           0.168444873           0.402869517
##       TeamGil Vicente         TeamGuimaraes         TeamMaritimo
##           0.434563727           0.495333115           0.250015382
##        TeamMoreirense     TeamPacos Ferreira      TeamPortimonense
##           0.091646781          -0.046992370           0.021141127
##             TeamPorto       TeamSanta Clara          TeamSp Braga
##           1.014211666           0.237137905           0.521506604
##         TeamSp Lisbon           TeamTondela           TeamVizela
##           0.851141140           0.330909485           0.215690420
##    OpponentBelenenses       OpponentBenfica      OpponentBoavista
##           0.009258891          -0.526027559          -0.025997711
##       OpponentEstoril     OpponentFamalicao    OpponentGil Vicente
##          -0.220562410          -0.037508877          -0.229735990
##     OpponentGuimaraes      OpponentMaritimo     OpponentMoreirense
##          -0.249948425          -0.193591503          -0.053387021
## OpponentPacos Ferreira OpponentPortimonense         OpponentPorto
##          -0.206593223          -0.181483213          -0.826533864
##    OpponentSanta Clara       OpponentSp Braga      OpponentSp Lisbon
##           0.010551500          -0.527780900          -0.799406984
##       OpponentTondela        OpponentVizela                  Home
##           0.231211729           0.080944900           0.171422266
```

Making predictions

- Sporting Lisbon is playing against Vizela at home

```
options(scipen=1, digits=4)
(lisbon_average <- predict(model, data.frame(Home=1, Team="Sp Lisbon", Opponent="Viz
```

```
##     1
## 2.919
```

- The same for Vizela

```
options(scipen=1, digits=4)
(vizela_average <- predict(model, data.frame(Home=0, Team="Vizela", Opponent="Sp Lis
        type="response"))
```

```
##      1
## 0.5402
```

When Sporting Lisbon is playing Vizela at Home, the expected (average) number of goals for Sporting Lisbon is 2.9195 and For Vizela is 0.5402

Let's do the calculations by hand for Sporting Lisbon

Calculations of λ for Sporting Lisbon

$$log(\lambda) = \beta_0 + \beta_{TeamSpLisbon} + \beta_{OpponentVizela} + \beta_{Home}$$

```
options(scipen=1, digits=4)
(intercept <- coefficients(model)["(Intercept)"])
```

```
## (Intercept)
##     -0.0321
```

```
(team_lisbon <- coefficients(model)["TeamSp Lisbon"])
```

```
## TeamSp Lisbon
##        0.8511
```

```
(oponnent_vizela <- coefficients(model)["OpponentVizela"])
```

```
## OpponentVizela
##        0.08094
```

```
(home_coeff <- coefficients(model)["Home"])
```

```
##   Home
## 0.1714
```

$log(\lambda) = 0.1714 + 0.8511 + 0.0809 + \text{-}0.0321 = 1.0714$

$\lambda = (1.0714) = 2.9195$

Making predictions: Home advantage

The average expected goals of the home team divided by the average expected goals by the same team playing away will be equal to exponent of Home coefficient

```
# lambda for Sporting Lisbon playing at home
spl_h=exp(intercept + team_lisbon + oponnent_vizela + home_coeff)
# lambda for Sporting Lisbon playing against Vizela away
vzl_a=exp(intercept + team_lisbon + oponnent_vizela)
spl_h/vzl_a
```

```
## (Intercept)
##       1.187
```

```
exp(home_coeff)
```

```
##  Home
## 1.187
```

Making predictions

As we have λ for both teams, we can use skellam distribution to come up with the probabilities.

```
# Lisbon's lambda
lisbon_average
```

```
##     1
## 2.919
```

```
# Vizela's lambda
vizela_average
```

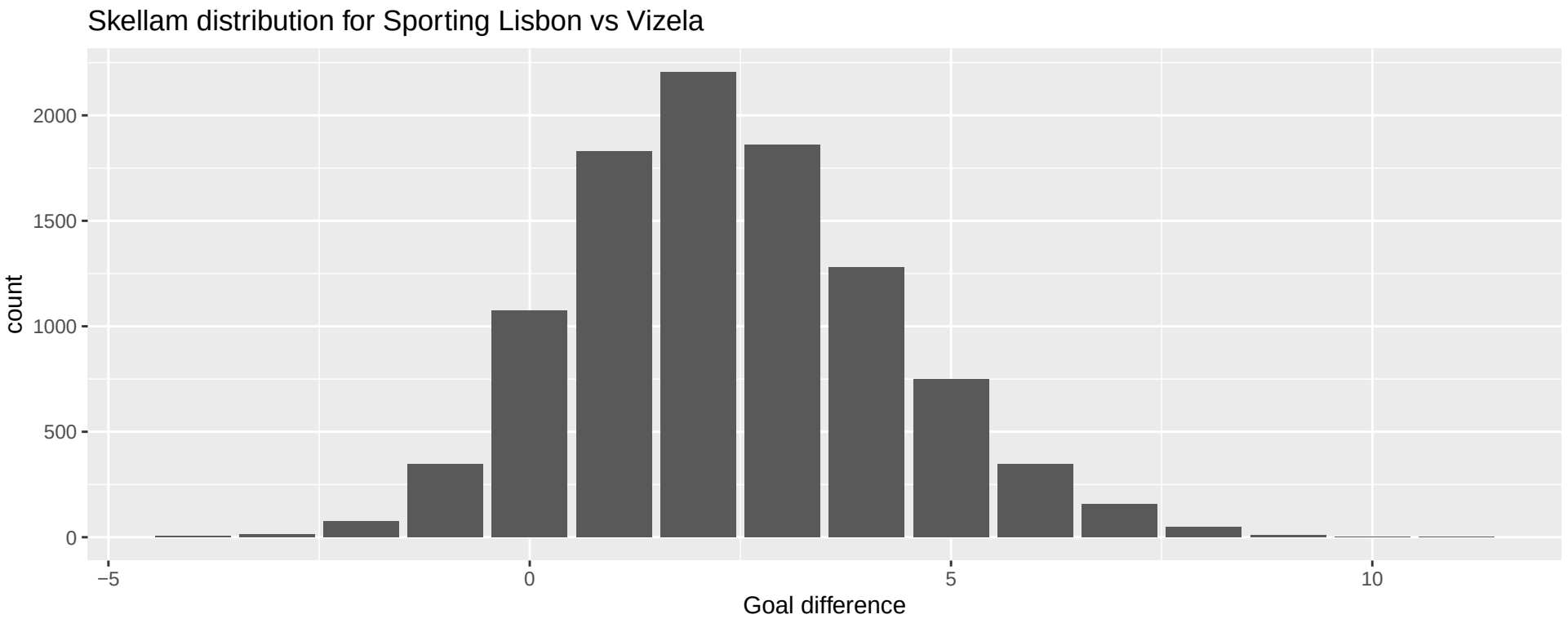
```
##      1
## 0.5402
```

```
# Simulate game results
set.seed(1)
k = rskellam(10000,mu1 = lisbon_average, mu2 = vizela_average)
```

Making predictions

A slight home team advantage

```
ggplot()+geom_bar(aes(x=k))+
  labs(x="Goal difference", title="Skellam distribution for Sporting Lisbon vs Vizel
```



Making predictions

Calculate the game output probabilities

```
# Away team to win
sum(dskellam(c(-100:-1),lisbon_average, vizela_average))
```

```
## [1] 0.0433
```

```
# Draw
sum(dskellam(0 ,lisbon_average, vizela_average))
```

```
## [1] 0.1044
```

```
# Home team to win
sum(dskellam(c(1:100),lisbon_average, vizela_average))
```

```
## [1] 0.8523
```

Predicting with models

Bradley-Terry model

Bradley-Terry model

The Bradley-Terry model (Bradley and Terry 1952) assumes that in a 'contest' between any two 'players', say player i and player j ($i, j \in \{1, \ldots . , K\}$) the ,odds that i beats j are α_i/α_j , where α_i and α_j are positive-valued parameters which represent the "ability" of each player/team.

The general model definition

$$P(\text{i beats j}) = \frac{\alpha_i}{\alpha_i+\alpha_j}$$

Sometimes the modification with exponential score function

$$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i}+e^{\alpha_j}}$$

Bradley-Terry model

$P(\text{i beats j}) = \frac{\alpha_i}{\alpha_i+\alpha_j}$

$logit(P(\text{i beats j})) = \log(\frac{P(\text{i beats j})}{P(\text{j beats i})}) = \log(\frac{P(\text{i beats j})}{1-P(\text{i beats j})}) = log(\frac{\alpha_i}{\alpha_i+\alpha_j} * \frac{\alpha_i+\alpha_j}{\alpha_j}) =$
$log(\alpha_i) - log(\alpha_j) = \lambda_i - \lambda_j$

The goal of the modeling is to find the best estimate for λ_i, λ_j

Bradley-Terry model

The modification with exponential score function

$$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

$$logit(P(\text{i beats j})) = \log(\frac{P(\text{i beats j})}{P(\text{j beats i})}) = \log(\frac{P(\text{i beats j})}{1 - P(\text{i beats j})}) = log(\frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}} * \frac{e^{\alpha_i} + e^{\alpha_j}}{e^{\alpha_j}}) = \alpha_i - \alpha_j$$

logit - log of odds

Bradley-Terry model

The basic Bradley-Terry model does not predict draw/tie games. You can use Davidson's extension on Bradley-Terry model if want to predict draw:
https://www.jstor.org/stable/2283595?seq=1#page_scan_tab_contents

Bradley-Terry model

```
data(nfl)
```

```
head(nfl, n=5)
```

```
##   schedule_season schedule_week schedule_date           team_away score_away
## 1            2000             1    2000-09-03  San Francisco 49ers         28
## 2            2000             1    2000-09-03     Tennessee Titans         13
## 3            2000             1    2000-09-03 Jacksonville Jaguars         27
## 4            2000             1    2000-09-03  Philadelphia Eagles         41
## 5            2000             1    2000-09-03        New York Jets         20
##           team_home score_home              stadium stadium_type
## 1   Atlanta Falcons         36         Georgia Dome       indoor
## 2     Buffalo Bills         16 Ralph Wilson Stadium      outdoor
## 3  Cleveland Browns          7  FirstEnergy Stadium      outdoor
## 4    Dallas Cowboys         14        Texas Stadium      outdoor
## 5 Green Bay Packers         16        Lambeau Field      outdoor
##   stadium_location result spread home_odds_open away_odds_open home_odds_close
## 1      Atlanta, GA      H     -8             NA             NA              NA
## 2 Orchard Park, NY      H     -3             NA             NA              NA
## 3    Cleveland, OH      A     20             NA             NA              NA
## 4       Irving, TX      A     27             NA             NA              NA
## 5    Green Bay, WI      A      4             NA             NA              NA
##   away_odds_close
## 1              NA
## 2              NA
## 3              NA
## 4              NA
## 5              NA
```

Bradley-Terry model

- Data on NFL matches
- Description
- Format
- A data frame with 4848 rows and 16 variables
- schedule_season: from 2000-2016
- schedule_week: The number of the week when the game was played
- schedule_date: The actual date of the game
- team_away: Away team
- score_away: The score of the away team
- team_home: Home team
- score_home: The score of home team
- stadium: The name of the stadium
- stadium_type: indoor, outdoor, retractable
- stadium_location: The location of the stadium: City and state abbreviation
- odds: closing and opening betting odds
- result: The result of the game: H=Home Win, T=Tie, A=Away Win
- spread: points spread, the difference between away team and home team scores ,
- home_odds_open, home_odds_close: Opening and closing decimal odds for the home team
- home_odds_open, away_odds_close: Opening and closing decimal odds for the away team

Bradley-Terry model

Some data preparation

```
nfl2 = nfl %>%
  select(team_home, team_away, result) %>%
  filter(result !="T") %>%
  mutate(ht_w = ifelse(result=="H",1,0),
         at_w = ifelse(result=="A",1,0))
head(nfl2)
```

```
##            team_home            team_away result ht_w at_w
## 1    Atlanta Falcons  San Francisco 49ers      H    1    0
## 2      Buffalo Bills     Tennessee Titans      H    1    0
## 3   Cleveland Browns Jacksonville Jaguars      A    0    1
## 4     Dallas Cowboys  Philadelphia Eagles      A    0    1
## 5  Green Bay Packers        New York Jets      A    0    1
## 6 Kansas City Chiefs   Indianapolis Colts      A    0    1
```

Bradley-Terry model

The final dataset

```
nfl3 = nfl2 %>%
    mutate(team_home=as.factor(team_home),
           team_away = as.factor(team_away)) %>%
    group_by(team_home, team_away) %>%
    summarise(ht=sum(ht_w), at=sum(at_w))

head(nfl3, n=5)
```

```
## # A tibble: 5 x 4
## # Groups:   team_home [1]
##   team_home          team_away            ht    at
##   <fct>              <fct>             <dbl> <dbl>
## 1 Arizona Cardinals Atlanta Falcons       3     1
## 2 Arizona Cardinals Baltimore Ravens      1     2
## 3 Arizona Cardinals Buffalo Bills         2     1
## 4 Arizona Cardinals Carolina Panthers     2     6
## 5 Arizona Cardinals Chicago Bears         0     3
```

Bradley-Terry model

Fitting the bradley-terry model

```
install.packages("BradleyTerry2")
```

```
model = BTm(cbind(ht, at), team_home, team_away, data=nfl3, id="team_")
```

Bradley-Terry model

Coefficients represent the teams abilities to win

```
summary(model)
```

```
##
## Call:
## BTm(outcome = cbind(ht, at), player1 = team_home, player2 = team_away,
##     id = "team_", data = nfl3)
##
## Deviance Residuals:
##    Min      1Q  Median      3Q     Max
## -2.840  -0.330   0.255   1.096   3.200
##
## Coefficients:
##                            Estimate Std. Error z value Pr(>|z|)
## team_Atlanta Falcons        0.23811    0.15249    1.56  0.11841
## team_Baltimore Ravens       0.61626    0.15754    3.91  9.2e-05 ***
## team_Buffalo Bills          0.14543    0.15662    0.93  0.35313
## team_Carolina Panthers      0.14710    0.15163    0.97  0.33197
## team_Chicago Bears          0.17298    0.15346    1.13  0.25966
## team_Cincinnati Bengals     0.11043    0.15705    0.70  0.48196
## team_Cleveland Browns      -0.34874    0.15968   -2.18  0.02896 *
## team_Dallas Cowboys         0.34812    0.15176    2.29  0.02180 *
## team_Denver Broncos         0.46813    0.15603    3.00  0.00270 **
## team_Detroit Lions         -0.34354    0.15529   -2.21  0.02695 *
## team_Green Bay Packers      0.74351    0.15606    4.76  1.9e-06 ***
## team_Houston Texans        -0.00846    0.16056   -0.05  0.95797
## team_Indianapolis Colts     0.69012    0.15798    4.37  1.3e-05 ***
## team_Jacksonville Jaguars  -0.24616    0.15807   -1.56  0.11940
## team_Kansas City Chiefs     0.42753    0.15594    2.74  0.00611 **
## team_Las Vegas Raiders     -0.08336    0.15659   -0.53  0.59450
## team_Los Angeles Chargers   0.28407    0.15555    1.83  0.06783 .
## team_Los Angeles Rams       0.11998    0.14480    0.83  0.40732
## team_Miami Dolphins         0.23783    0.15645    1.52  0.12847
## team_Minnesota Vikings      0.27640    0.15293    1.81  0.07070 .
## team_New England Patriots   1.18548    0.16429    7.22  5.4e-13 ***
```

Bradley-Terry model

New England Patriots: the strongest team Cleveland Browns: the weakest team

```
coef=model$coefficients
sort(coef, decreasing = T)
```

```
##  team_New England Patriots   team_Pittsburgh Steelers
##                   1.185479                   0.831506
##     team_Green Bay Packers    team_Indianapolis Colts
##                   0.743509                   0.690121
##      team_Baltimore Ravens    team_New Orleans Saints
##                   0.616258                   0.526252
##   team_Philadelphia Eagles      team_Seattle Seahawks
##                   0.513035                   0.479009
##        team_Denver Broncos    team_Kansas City Chiefs
##                   0.468134                   0.427533
##        team_Dallas Cowboys      team_Tennessee Titans
##                   0.348124                   0.299783
##  team_Los Angeles Chargers     team_Minnesota Vikings
##                   0.284065                   0.276397
##       team_Atlanta Falcons        team_Miami Dolphins
##                   0.238111                   0.237828
##         team_Chicago Bears     team_Carolina Panthers
##                   0.172982                   0.147104
##         team_Buffalo Bills       team_New York Giants
##                   0.145429                   0.137314
##      team_Los Angeles Rams    team_Cincinnati Bengals
##                   0.119982                   0.110432
##   team_San Francisco 49ers  team_Tampa Bay Buccaneers
##                   0.103724                   0.086271
##         team_New York Jets        team_Houston Texans
##                   0.069364                  -0.008462
## team_Washington Commanders     team_Las Vegas Raiders
##                  -0.075417                  -0.083359
##  team_Jacksonville Jaguars         team_Detroit Lions
##                  -0.246156                  -0.343538
##      team_Cleveland Browns
```

Bradley-Terry model

Overall there are 32 teams in the data

```
length(unique(nfl3$team_home))
```

```
## [1] 32
```

```
# and we have coefficients for 31
length(coef)
```

```
## [1] 31
```

- One team is taken as a reference level
- Its ability is $= 0$
- In our case that is Arizona Cardinals as it comes first alphabetically

Bradley-Terry model

- Abilities are calculated relative to Arizona Cardinales
- Thus, positive ability means that the team is doing better than Cardinals
- Negative ability is showing that the team is doing worst than Cardinals

```
BTabilities(model)
```

```
##                          ability   s.e.
## Arizona Cardinals       0.000000 0.0000
## Atlanta Falcons         0.238111 0.1525
## Baltimore Ravens        0.616258 0.1575
## Buffalo Bills           0.145429 0.1566
## Carolina Panthers       0.147104 0.1516
## Chicago Bears           0.172982 0.1535
## Cincinnati Bengals      0.110432 0.1571
## Cleveland Browns       -0.348735 0.1597
## Dallas Cowboys          0.348124 0.1518
## Denver Broncos          0.468134 0.1560
## Detroit Lions          -0.343538 0.1553
## Green Bay Packers       0.743509 0.1561
## Houston Texans         -0.008462 0.1606
## Indianapolis Colts      0.690121 0.1580
## Jacksonville Jaguars   -0.246156 0.1581
## Kansas City Chiefs      0.427533 0.1559
## Las Vegas Raiders      -0.083359 0.1566
```

Bradley-Terry model

- Let's do some plotting
- First let's create a dataframe with abilities

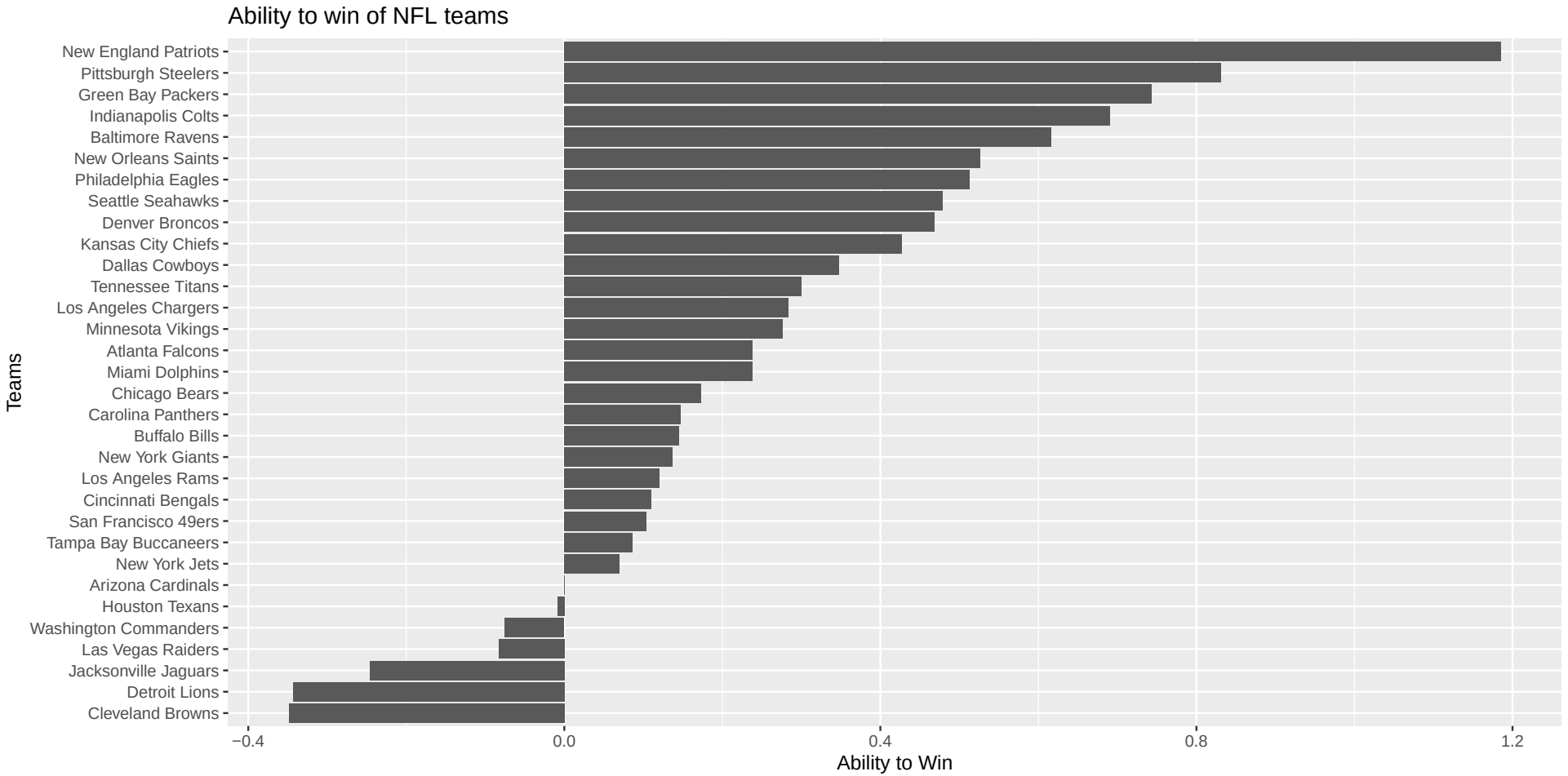
```
abilities = as.data.frame(BTabilities(model))
abilities$team = rownames(abilities)
abilities = abilities[order(abilities$ability, decreasing = T),]
abilities
```

```
##                          ability   s.e.                    team
## New England Patriots   1.185479 0.1643  New England Patriots
## Pittsburgh Steelers    0.831506 0.1600   Pittsburgh Steelers
## Green Bay Packers      0.743509 0.1561     Green Bay Packers
## Indianapolis Colts     0.690121 0.1580    Indianapolis Colts
## Baltimore Ravens       0.616258 0.1575      Baltimore Ravens
## New Orleans Saints     0.526252 0.1538    New Orleans Saints
## Philadelphia Eagles    0.513035 0.1527   Philadelphia Eagles
## Seattle Seahawks       0.479009 0.1458      Seattle Seahawks
## Denver Broncos         0.468134 0.1560        Denver Broncos
## Kansas City Chiefs     0.427533 0.1559    Kansas City Chiefs
## Dallas Cowboys         0.348124 0.1518        Dallas Cowboys
## Tennessee Titans       0.299783 0.1562      Tennessee Titans
## Los Angeles Chargers   0.284065 0.1556  Los Angeles Chargers
## Minnesota Vikings      0.276397 0.1529     Minnesota Vikings
## Atlanta Falcons        0.238111 0.1525       Atlanta Falcons
```

Bradley-Terry model

Abilities from strongest to weakest

```
ggplot(data=abilities, aes(x = reorder(team, ability), y = ability)) +
  geom_bar(stat="identity") + coord_flip() +
  labs(x="Teams", y="Ability to Win", title="Ability to win of NFL teams")
```



Bradley-Terry model

everything relative to the worst team (Cleveland Browns)

- Let's say we want to calculate the abilities relative to the weakest team, so all the abilities would be ≥ 0
- You could do this by re leveling the team variables

```
nfl4=nfl2 %>%
    mutate(team_home = relevel(team_home, ref="Cleveland Browns"),
           team_away = relevel(team_away, ref="Cleveland Browns")) %>%
    group_by(team_home, team_away) %>% summarise(ht=sum(ht_w), at=sum(at_w))
model_browns = BTm(cbind(ht, at), team_home, team_away, data=nfl4, id="team_")
```

OR Standardize the variable ability around its minimum (these two approaches will give exactly the same result)

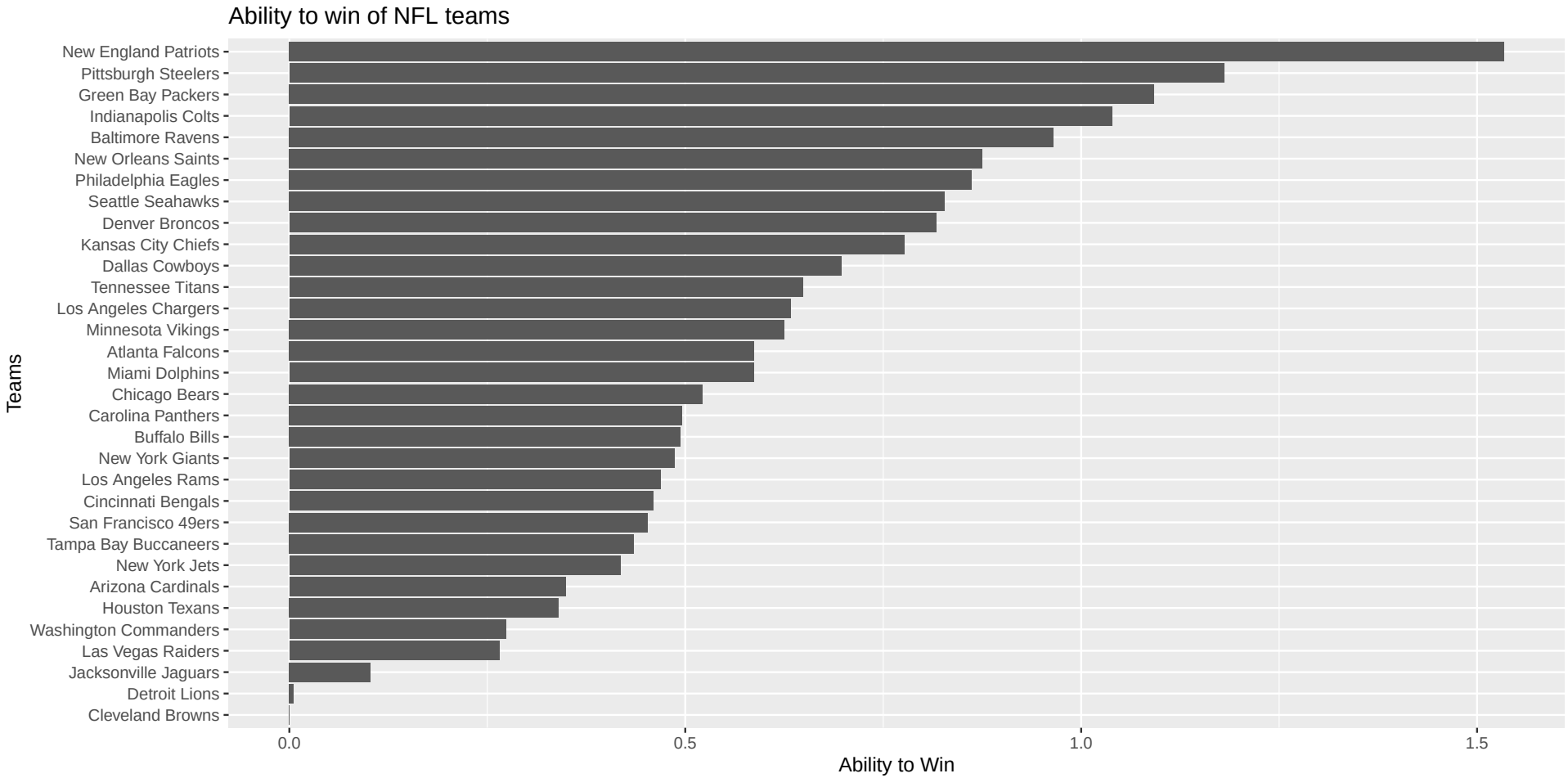
```
abilities_brown = abilities
abilities_brown$ability = abilities_brown$ability - min(abilities$ability)
head(abilities_brown)
```

```
##                      ability   s.e.                team
## New England Patriots   1.534 0.1643 New England Patriots
## Pittsburgh Steelers    1.180 0.1600  Pittsburgh Steelers
## Green Bay Packers      1.092 0.1561    Green Bay Packers
## Indianapolis Colts     1.039 0.1580   Indianapolis Colts
## Baltimore Ravens       0.965 0.1575     Baltimore Ravens
## New Orleans Saints     0.875 0.1538   New Orleans Saints
```

Bradley-Terry model

Abilities plot

```
ggplot(data=abilities_brown, aes(x=reorder(team,ability), y=ability)) +
  geom_bar(stat="identity") + coord_flip() +
  labs(x="Teams", y="Ability to Win", title="Ability to win of NFL teams")
```



Bradley-Terry model

Let's do predictions for a sample of games for Denver Broncos

```
broncos=data.frame(team_home=rep("Denver Broncos", 4),
                   team_away=c("Cleveland Browns", "Tennessee Titans",
                               "Chicago Bears", "Detroit Lions"))
# we need to make team_home and team_away having the same factor structure
# as in the initial model

broncos$team_home=factor(broncos$team_home,
                         levels(nfl3$team_away))

broncos$team_away=factor(broncos$team_away,
                         levels(nfl3$team_away))
str(broncos)
```

```
## 'data.frame':    4 obs. of  2 variables:
##  $ team_home: Factor w/ 32 levels "Arizona Cardinals",..: 10 10 10 10
##  $ team_away: Factor w/ 32 levels "Arizona Cardinals",..: 8 31 6 11
```

```
broncos
```

```
##        team_home        team_away
## 1 Denver Broncos Cleveland Browns
## 2 Denver Broncos Tennessee Titans
## 3 Denver Broncos    Chicago Bears
## 4 Denver Broncos    Detroit Lions
```

Bradley-Terry model

This is the probability of Denver Broncos winning

```
( broncos_prob = predict(model, newdata=broncos, level=2, type="response") )
```

```
##      1      2      3      4
## 0.6936 0.5420 0.5733 0.6925
```

Make a dataframe with probs

```
( broncos_df = data.frame(broncos, ht_w=broncos_prob,
                                   at_w=1-broncos_prob) )
```

```
##        team_home        team_away   ht_w   at_w
## 1 Denver Broncos Cleveland Browns 0.6936 0.3064
## 2 Denver Broncos Tennessee Titans 0.5420 0.4580
## 3 Denver Broncos    Chicago Bears 0.5733 0.4267
## 4 Denver Broncos    Detroit Lions 0.6925 0.3075
```

Bradley-Terry model

Probabilities are predicted using the following formula

$$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$$

$$P(\text{j beats i}) = \frac{e^{\alpha_j}}{e^{\alpha_i} + e^{\alpha_j}}$$

Bradley-Terry model

Calculating probabilities manually for Denver Broncos-Cleveland Browns

```
#first get abbilities
broncos_ab = abilities[rownames(abilities)=="Denver Broncos",1]
browns_ab = abilities[rownames(abilities)=="Cleveland Browns",1]

broncos_ab
```

```
## [1] 0.4681
```

```
browns_ab
```

```
## [1] -0.3487
```

```
# probability of Broncos to win
exp(broncos_ab)/(exp(broncos_ab)+exp(browns_ab))
```

```
## [1] 0.6936
```

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i}+e^{\alpha_j}}$

Bradley-Terry model with order effect

Bradley-Terry model with order effect

Bradley-Terry model with order effect

- In certain types of application some or all contests have an associated 'bias', related to the order in which items are presented to a judge or with the location in which a contest takes place, for example.
- In sports this can be related to home advantage: the ability of the team is different when it plays at home compared to when it plays away.
- In this case the model has the following structure

z = 1 if i has the supposed advantage and z = -1 if j has it z = 0 if there is no advantage

Bradley-Terry model with order effect

We need to do several data transformations to get the model to account home advantage factor.

```
nfl3$team_home = data.frame(team = nfl3$team_home, at.home = 1)
nfl3$team_away = data.frame(team = nfl3$team_away, at.home = 0)
```

The result is a grouped_df

```
str(nfl3)
```

```
## grouped_df [992 x 4] (S3: grouped_df/tbl_df/tbl/data.frame)
##  $ team_home:'data.frame':   992 obs. of  2 variables:
##   ..$ team   : Factor w/ 32 levels "Arizona Cardinals",..: 1 1 1 1 1 1 1 1 1 1 ...
##   ..$ at.home: num [1:992] 1 1 1 1 1 1 1 1 1 1 ...
##  $ team_away:'data.frame':   992 obs. of  2 variables:
##   ..$ team   : Factor w/ 32 levels "Arizona Cardinals",..: 2 3 4 5 6 7 8 9 10 11 ...
##   ..$ at.home: num [1:992] 0 0 0 0 0 0 0 0 0 0 ...
##  $ ht       : num [1:992] 3 1 2 2 0 2 4 6 1 7 ...
##  $ at       : num [1:992] 1 2 1 6 3 0 0 2 3 2 ...
##  - attr(*, "groups")= tibble [32 x 2] (S3: tbl_df/tbl/data.frame)
##   ..$ team_home:'data.frame':    32 obs. of  2 variables:
##   .. ..$ team   : Factor w/ 32 levels "Arizona Cardinals",..: 1 2 3 4 5 6 7 8 9 10 ...
##   .. ..$ at.home: num [1:32] 1 1 1 1 1 1 1 1 1 1 ...
##   ..$ .rows    : list<int> [1:32]
##   .. ..$ : int [1:31] 1 2 3 4 5 6 7 8 9 10 ...
##   .. ..$ : int [1:31] 32 33 34 35 36 37 38 39 40 41 ...
##   .. ..$ : int [1:31] 63 64 65 66 67 68 69 70 71 72 ...
##   .. ..$ : int [1:31] 94 95 96 97 98 99 100 101 102 103 ...
##   .. ..$ : int [1:31] 125 126 127 128 129 130 131 132 133 134 ...
##   .. ..$ : int [1:31] 156 157 158 159 160 161 162 163 164 165 ...
##   .. ..$ : int [1:31] 187 188 189 190 191 192 193 194 195 196 ...
##   .. ..$ : int [1:31] 218 219 220 221 222 223 224 225 226 227 ...
##   .. ..$ : int [1:31] 249 250 251 252 253 254 255 256 257 258 ...
##   .. ..$ : int [1:31] 280 281 282 283 284 285 286 287 288 289 ...
##   .. ..$ : int [1:31] 311 312 313 314 315 316 317 318 319 320
```

Bradley-Terry model with order effect

```
head(nfl3$team_home, n=5)
```

```
##                 team at.home
## 1 Arizona Cardinals       1
## 2 Arizona Cardinals       1
## 3 Arizona Cardinals       1
## 4 Arizona Cardinals       1
## 5 Arizona Cardinals       1
```

Bradley-Terry model with order effect

```
model2 = BTm(cbind(ht, at), team_home, team_away, formula=~team+at.home,
             data=nfl3, id="team")
```

```
model2$coefficients
```

```
##       teamAtlanta Falcons      teamBaltimore Ravens         teamBuffalo Bills
##                   0.24043                   0.62247                   0.14382
##     teamCarolina Panthers        teamChicago Bears    teamCincinnati Bengals
##                   0.14930                   0.17455                   0.10940
##      teamCleveland Browns        teamDallas Cowboys        teamDenver Broncos
##                  -0.35878                   0.35159                   0.47230
##         teamDetroit Lions     teamGreen Bay Packers        teamHouston Texans
##                  -0.35232                   0.75438                  -0.01339
##    teamIndianapolis Colts  teamJacksonville Jaguars    teamKansas City Chiefs
##                   0.69764                  -0.25483                   0.43084
##     teamLas Vegas Raiders  teamLos Angeles Chargers      teamLos Angeles Rams
##                  -0.08974                   0.28443                   0.11828
##        teamMiami Dolphins     teamMinnesota Vikings  teamNew England Patriots
##                   0.23734                   0.27772                   1.20034
##    teamNew Orleans Saints       teamNew York Giants        teamNew York Jets
##                   0.53564                   0.13660                   0.06604
##   teamPhiladelphia Eagles   teamPittsburgh Steelers   teamSan Francisco 49ers
##                   0.51983                   0.84212                   0.10597
##      teamSeattle Seahawks  teamTampa Bay Buccaneers      teamTennessee Titans
##                   0.48474                   0.08710                   0.30058
## teamWashington Commanders                   at.home
##                  -0.07918                   0.26589
```

Bradley-Terry model with order effect

Let make predictions for Broncos vs Browns by hand, assuming that Broncos are playing at home

```
( db = coefficients(model2)['teamDenver Broncos'] )
```

```
## teamDenver Broncos
##             0.4723
```

```
( cb = coefficients(model2)['teamCleveland Browns'] )
```

```
## teamCleveland Browns
##             -0.3588
```

```
(h = coefficients(model2)['at.home'])
```

```
## at.home
##  0.2659
```

Bradley-Terry model with order effect

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i} + e^{\alpha_j}}$

Probability of Broncos to win at home

```
( w_h = exp(db+h)/(exp(db+h)+exp(cb)) )
```

```
## teamDenver Broncos
##             0.7497
```

Odds of Broncos to win at home

```
( od_h = w_h/(1-w_h) )
```

```
## teamDenver Broncos
##              2.995
```

Bradley-Terry model with order effect

Playing at "neutral" field, no advantage for any team

$P(\text{i beats j}) = \frac{e^{\alpha_i}}{e^{\alpha_i}+e^{\alpha_j}}$

```
( w_n = exp(db)/(exp(db)+exp(cb)) )
```

```
## teamDenver Broncos
##             0.6966
```

Odds of wining at "neutral" field

```
( od_n = w_n/(1-w_n) )
```

```
## teamDenver Broncos
##              2.296
```

Bradley-Terry model with order effect

Odds ratio and home team advantage

```
od_h/od_n
```

```
## teamDenver Broncos
##              1.305
```

```
exp(coefficients(model2)['at.home'])
```

```
## at.home
##   1.305
```

$e^{\text{at.home}}$ is the odd multiplier for a home team to win compared to the neutral field

Bradley-Terry model with order effect

The same game, now Broncos is away, Browns is at home

$P(\text{j beats i}) = \frac{e^{\alpha_j + h}}{e^{\alpha_i} + e^{\alpha_j + h}}$

```
( w_a = exp(db-h)/(exp(db-h)+exp(cb)) )
```

```
## teamDenver Broncos
##             0.6377
```

Probability of Browns to win at home

```
1 - exp(db-h)/(exp(db-h)+exp(cb))
```

```
## teamDenver Broncos
##             0.3623
```

```
# OR
exp(cb+h)/(exp(db)+exp(cb+h))
```

```
## teamCleveland Browns
##               0.3623
```

Bradley-Terry model

Making predictions for the series of games

```
broncos = data.frame(team_home=rep("Denver Broncos", 4),
                     team_away=c("Cleveland Browns", "Tennessee Titans",
                                 "Chicago Bears", "Detroit Lions"))

broncos$team_home=data.frame(team = factor(broncos$team_home,
                                           levels(nfl3$team_home$team)),
                                           at.home = 1)
broncos$team_away=data.frame(team = factor(broncos$team_away,
                                           levels(nfl3$team_home$team)),
                                           at.home = 0)
broncos
```

```
##   team_home.team team_home.at.home   team_away.team team_away.at.home
## 1 Denver Broncos                 1 Cleveland Browns                 0
## 2 Denver Broncos                 1 Tennessee Titans                 0
## 3 Denver Broncos                 1    Chicago Bears                 0
## 4 Denver Broncos                 1    Detroit Lions                 0
```

Bradley-Terry model

```
( br2 = predict(model2, newdata=broncos, level=2, type="response") )
```

```
##      1      2      3      4
## 0.7497 0.6077 0.6373 0.7485
```